

AAA Concepts:

- o AAA is a term which stands for Authentication, Authorization and Accounting.
- o AAA is a centralized management of users to access the network devices etc.
- o AAA services allow setting up access control on Cisco Routers & Cisco Switches.
- o Whenever user's attempts to login these network devices verifies by AAA database.
- o User management done on AAA database without need to reconfigure each device.
- o AAA also control connections passing through router for access network resources.
- o AAA provides flexibility and scalability, using privilege levels allows the flexibility.
- o AAA server can be RADIUS protocol or TACACS+ protocol where database located.
- o Use the local database fallback if the TACACS+/RADIUS server becomes unavailable.
- o Fallback method only occurs and work if the AAA server is unavailable or down.
- o There are two uses of AAA one is device administration and second network access.

Authentication
Who are you?

Authorization
How much can you spend?

Accounting
What did you spend it on?

Credit Card Statement Details:

Account Number: 1234-567-890 | Statement Closing Date: 01-31-01 | Current Amount Due: \$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99900-1234

MAIL PAYMENT TO:
THE BANK
132 VINE STREET
ANYTOWN, USA 67500-0010

872919345 00178255000000003

Statement of Personal Credit Card Account

Cardmember Name: JOE EMPLOYEE | Account Number: 1234-456-890 | Statement Closing Date: 01-31-01

Statement Date: 02-01-01 | Payment Due Date: 03-01-01

Closing Date: 01-31-01

Credit Limit: \$1,500.00 | Credit Available: \$1221.50

New Balance: \$278.50 | Minimum Payment Due: \$20.00

Account Summary

Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	\$278.50

Reference Number	Sold	Posted	Activity Since Last Statement	Amount
43210987	01-03	01-13	Payment, Thank You	-\$74.25
01234567	01-12	01-13	Wings 'N' Things Anytown, USA	\$25.25
78901234	01-14	01-17	Record Release Anytown, USA	\$40.00
45678901	01-14	01-17	Sports Stadium Anytown, USA	\$75.25
3210987	01-22	01-23	Tie Tack Anytown, USA	\$20.75
78543210	01-29	01-30	Electronic World Anytown, USA	\$89.25
2345678		01-30	Transaction Fees	\$3.00
34567890		01-01	Annual Fee	\$25.00

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Authentication:

- o Check the user's credentials to verify their identity.
- o Authentication is the process of proving identity to the system.
- o Authentication is the validation of an identity or a credential.
- o The user or computer has to prove its identity to the server or client.
- o Log on to a PC with a user name and password you are authenticating.
- o Authentication does not determine what tasks the individual can do.
- o Authentication merely identifies and verifies who the person or system is.
- o In short or summary basically, Authentication is about who somebody is.

Authorization:

- o Authorization determines what the user is allowed to do in network.
- o Authorization restrict the access to the user when login to device.
- o Server determines client has permission to use a resource or access file.
- o Authorization is the process of verifying access to something.
- o In short or summary Authorization is about what they are allowed to do.
- o Authorization is the function of specifying access rights to resources.
- o Authorization is process to confirm what you are authorized to perform.

Accounting:

- o In AAA model, accounting features is also very much important in security.
- o Accounting command tracking commands, services and resources used by user.
- o Accounting is the measure of resources consumed by a user during access.
- o Accounting, includes amount of time, amount of data user has send or received.
- o Accounting is carried in the form of logs of session statistics and usage information.
- o Accounting data is used for authorization control, resources utilization, billing & planning.
- o Accounting is also very much helpful to troubleshoot if network devices are not working.
- o Accounting is disabled by default in Authentication, Authorization and Accounting model.
- o AAA Accounting Types Network, Exec, Commands, Connection and Resource.
- o Accounting collects and stores information regarding and about logins and activity.
- o Accounting recorded while accessing the specific device during specific time.

AAA Options:

- o Cisco provides a number of ways to implement AAA.
- o Two main protocols are TACACS Server and Radius Server.

AAA with TACACS+:

- o TACACS+ stands for Terminal Access Controller Access Control System Plus.
- o TACACS+ is a Cisco proprietary protocol that is use to deliver AAA security services.
- o TACACS+ is an application, which is implement through AAA.
- o TACACS+ provides centralized acceptance of user to take the access control of routers.
- o TACACS+ provides other access servers in the network as well.
- o TACACS+ provides to control the authorization of router commands per-user or group.
- o TACACS+ offers multiprotocol support.
- o TACACS+ encrypts entire body of the packet but leaves a standard TACACS+ header.
- o TACACS+ uses the AAA architecture, which separates AAA.

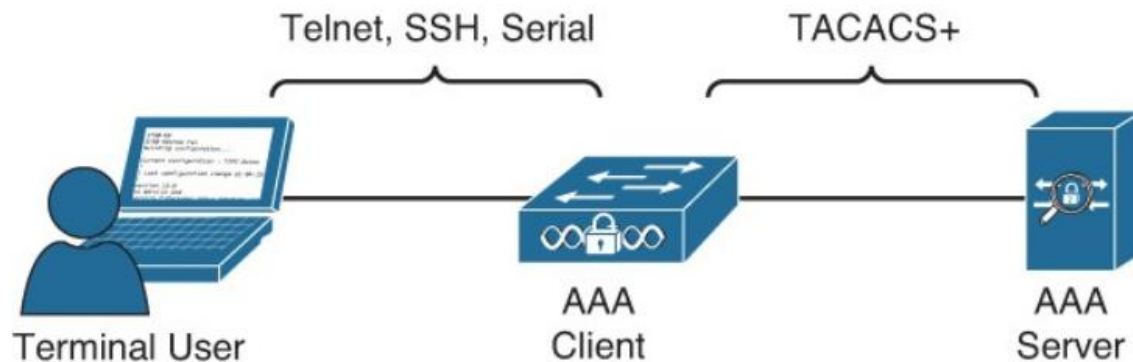
AAA with RADIUS:

- o RADIUS stand for Remote Authentication Dial in User Service.
- o RADIUS is a security protocol that secures the network against unauthorized access.
- o RADIUS clients run on routers & send authentication request to a centralized server.
- o RADIUS Server contains network service access information and user authentication.
- o RADIUS does not allow users to control which commands can be executed or not.
- o RADIUS is not as useful for router or switch management.
- o RADIUS does not support multiprotocol.
- o RADIUS encrypts password of the access-request packet only from Client to server.
- o RADIUS uses UDP as a transport protocol while TACACS+ uses TCP.
- o RADIUS combines authentication and authorization processes.

RADIUS	TACACS+
RADIUS uses UDP	TACACS+ uses TCP
Uses ports 1812/1645 for authentication Uses ports 1813/1646 for accounting	TACACS+ uses TCP port 49
RADIUS encrypts passwords only	TACACS+ encrypts the entire communication
RADIUS combines authentication and Authorization	TACACS+ treats Authentication, Authorization, and Accountability differently
RADIUS is an open protocol	TACACS+ is Cisco proprietary protocol
RADIUS is a light-weight protocol consuming less resources	TACACS+ is a heavy-weight protocol consuming more resources
RADIUS is limited to privilege mode	TACACS+ supports 15 privilege levels
Mainly used for Network Access	Mainly used for Device Administration

Device Administration:

- o Controlling access to who can log in to a network device console, telnet, SSH session.
- o Controlling access to who can log in to a network device via other methods.
- o Device administration is a process of AAA for controlling the access to network device.
- o Which can be by any methods via Telnet session, VTY, TTY, SSH session or via Console.
- o Device Administrator is user logs into the network devices such as switches routers etc.
- o In order to perform the configuration and maintenance of the administered devices.
- o There are two uses of AAA, Device Administration and Network Access.



Network Access:

- o Securing network access can provide the identity of the device or user.
- o Secure network access is necessary in order to identify the user or endpoint.
- o Before permitting the entity to communicate or access the network.
- o AAA has important role in Network Access authentication and authorization.
- o To filter legitimate user AAA Network access authentication is required.
- o AAA authenticates these devices & control what these users are authorized for.
- o There are two uses of AAA, Device Administration and Network Access.



Local Privilege Authorization Fallback:

- o For several functions local database act as fallback method.
- o It is design to help the user to prevent accidental lockout from security devices.
- o Use local database fallback if the TACACS+/RADIUS server becomes unavailable.
- o Fallback only occurs if the AAA Server is unavailable or down.